Unit I :: Distribution of Velocities and Molecular Collisions

PHY0500304: Heat & Thermodynamics

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Syllabus

Unit I: Distribution of Velocities and Molecular Collisions

Maxwell-Boltzmann Law of Distribution of Velocities in an Ideal Gas and its Experimental Verification. Mean, RMS and Most Probable Speeds. Degrees of Freedom. Law of Equipartition of Energy (No proof required). Mean Free Path. Collision Probability. Transport Phenomenon in Ideal Gases: (1) Viscosity, and (2) Thermal Conductivity. Brownian Motion (qualitative idea only).

1 Maxwell-Boltzmann distribution law

The Maxwell-Boltzmann distribution is a result of the kinetic theory of gases, which provides a simplified explanation of many fundamental gaseous properties, including pressure and diffusion. This distribution applies fundamentally to particle velocities in three dimensions, but turns out to depend only on the speed (the magnitude of the velocity) of the particles. A particle speed probability distribution indicates which speeds are more likely: a randomly chosen particle will have a speed selected randomly from the distribution, and is more likely to be within one range of speeds than another.

The kinetic theory of gases applies to the classical ideal gas, which is an idealization of real gases. In real gases, there are various effects (e.g., van der Waals interactions, vortical flow, relativistic speed limits, and quantum exchange interactions) that can make their speed distribution different from the Maxwell–Boltzmann form. However, rarefied gases at ordinary temperatures behave very nearly like an ideal gas and the Maxwell speed distribution is an excellent approximation for such gases. This is also true for ideal plasmas, which are ionized gases of sufficiently low density.

1.1 Postulates of kinetic theory of gases

1. Every mole of a gas contains about 6.023×10^{23} molecules.
2. The gas molecules may be regarded as point masses.
3. The gas molecules are in a state of continuous random motion.
4. In the absence of an external force field, the molecules are distributed uniformly in the container, i.e. an ideal gas behaves as an isotropic medium.
5. The molecules of a gas experience force only during collisions.
6. The molecules of a gas behave as rigid, perfectly elastic hard spheres.
7. The duration of collisions is negligible compared to the time interval between collisions.
8. There is a spread of molecular speeds from zero to infinity.
9. The gravitational potential energy has negligible effect on the motion of the gas molecules.

1.2 Fundamental assumptions

In the derivation of the Maxwell-Boltzmann distribution, we make the following basic assumptions:

1. In the equilibrium state, the molecules have complete randomness of direction and velocity.
2. There is no mass motion or convection current in the body of the gas.

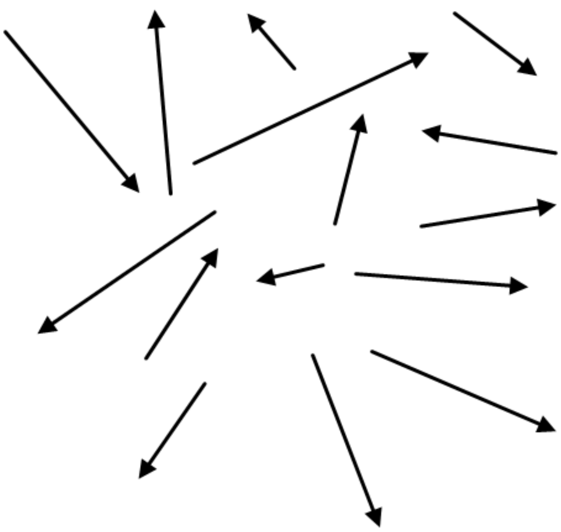
3. The probability of a molecule having an x-velocity component at any point of time is independent of its y- and z-velocity components, and so on.

4. The probability that a molecule selected at random has velocity component in a given range is a function only of the magnitude of the velocity component and the width of the interval.

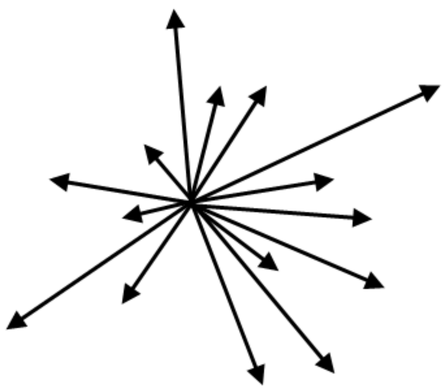
5. The gas molecules have no rotational or vibrational energies.

1.3 Velocity space

Consider a gas of N molecules enclosed in a vessel, of arbitrary shape, and moving randomly. The velocity vectors are all randomly oriented in all possible directions.



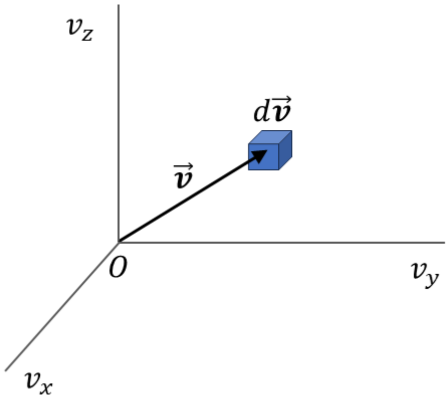
Now we transport each vector parallel to itself to a common origin.



If we now fix the velocity axes v_x, v_y and v_z with respect to a common origin O (that corresponds to particles at rest), every molecular velocity can be represented in this space, which is known as the *velocity space*.

1.4 Derivation of Maxwell-Boltzmann distribution law

Let us now consider an infinitessimal volume element $\mathrm{d}\vec{\boldsymbol{v}}$ in this velocity space. The number of velocity vectors that terminate inside $\mathrm{d}\vec{\boldsymbol{v}}$ gives us the average number of molecules whose velocities lie between $\vec{\boldsymbol{v}}$ and $\vec{\boldsymbol{v}}+\mathrm{d}\vec{\boldsymbol{v}}$.



We have, for any velocity $\vec{\boldsymbol{v}}$, the corresponding speed to be

$$v^2 = v_x^2 + v_y^2 + v_z^2 \tag{1}$$

Let $\mathrm{d}N_{v_x}$ be the number of molecules having velocity components in the range v_x and $v_x + \mathrm{d}v_x$. Then from assumption #4, we have

$$\frac{\mathrm{d}N_{v_x}}{N} = f(v_x)\,\mathrm{d}v_x$$
$$\text{or, } \mathrm{d}N_{v_x} = N f(v_x)\,\mathrm{d}v_x \tag{2}$$

In equation (2), $f(v_x)$ is an unknown function that we have to determine.

Also, the probability of finding a molecule with x-velocity component in the range v_x and $v_x+\mathrm{d}v_x$ is

$$p_x = \frac{\mathrm{d}N_{v_x}}{N} = f(v_x)\,\mathrm{d}v_x \tag{3}$$

Likewise, from assumption #3, we have

$$p_y = f(v_y)\,\mathrm{d}v_y \tag{4}$$
$$p_z = f(v_y)\,\mathrm{d}v_z \tag{5}$$

Then from the law of compound probabilities, the probability of finding a molecule having velocity components simultaneously in the ranges $[v_x, v_x + \mathrm{d}v_x]$, $[v_y, v_y + \mathrm{d}v_y]$ and $[v_z, v_z + \mathrm{d}v_z]$ is

$$\frac{\mathrm{d}^3 N_{v_x v_y v_z}}{N} = p_x p_y p_z = f(v_x) f(v_y) f(v_z)\,\mathrm{d}v_x\,\mathrm{d}v_y\,\mathrm{d}v_z \tag{6}$$

Hence, the number density of velocity points is

$$\rho = \frac{\mathrm{d}^3 N_{v_x v_y v_z}}{\mathrm{d}v_x\,\mathrm{d}v_y\,\mathrm{d}v_z} = N f(v_x) f(v_y) f(v_z) \tag{7}$$

It follows from assumption #1 that the velocity space has been assumed to be isotropic, and so this implies that ρ is independent of the direction of $\vec{v}$ and only depends ton v^2, i.e

$$\begin{aligned} \rho &\propto J(v^2) \\ \implies f(v_x)f(v_y)f(v_z) &= J(v^2) \end{aligned} \tag{8}$$

where J is some other function.

For a fixed value of v^2, $J(v^2)$ is constant and so

$$\begin{aligned} \mathrm{d}\big[J(v^2)\big] &= 0 \\ \implies \mathrm{d}\big[f(v_x)f(v_y)f(v_z)\big] &= 0 \\ \implies \frac{\partial f(v_x)}{\partial v_x}\mathrm{d}v_x\, f(v_y)f(v_z) + f(v_x)\frac{\partial f(v_y)}{\partial v_y}\mathrm{d}v_y\, f(v_z) + f(v_x)f(v_y)\frac{\partial f(v_z)}{\partial v_z}\mathrm{d}v_z &= 0 \end{aligned} \tag{9}$$

Dividing equation (9) by $f(v_x)f(v_y)f(v_z)$, we obtain

$$\frac{1}{f(v_x)}\frac{\partial f(v_x)}{\partial v_x}\mathrm{d}v_x + \frac{1}{f(v_y)}\frac{\partial f(v_y)}{\partial v_y}\mathrm{d}v_y + \frac{1}{f(v_z)}\frac{\partial f(v_z)}{\partial v_z}\mathrm{d}v_z = 0 \tag{10}$$

Also, for a fixed value of v^2, we have

$$\begin{aligned} \mathrm{d}\big[v^2\big] &= 0 \\ \implies \mathrm{d}\big[v_x^2 + v_y^2 + v_z^2\big] &= 0 \\ \implies v_x\,\mathrm{d}v_x + v_y\,\mathrm{d}v_y + v_z\,\mathrm{d}v_z &= 0 \end{aligned} \tag{11}$$

Suppose that B is some unknown constant. Then multiplying equation (11) with $2B$ and adding to equation (10) gives

$$\left[\frac{1}{f(v_x)}\frac{\partial f(v_x)}{\partial v_x} + 2Bv_x\right]\mathrm{d}v_x + \left[\frac{1}{f(v_y)}\frac{\partial f(v_y)}{\partial v_y} + 2Bv_y\right]\mathrm{d}v_y + \left[\frac{1}{f(v_z)}\frac{\partial f(v_z)}{\partial v_z} + 2Bv_z\right]\mathrm{d}v_z = 0 \tag{12}$$

Even though v is held constant, v_x, v_y, v_z may individually change and yet satisfy equation (1). In equation (12), the first term within brackets depends only on v_x, the second term on v_y and the third term on v_z.

Because $\mathrm{d}v_x$, $\mathrm{d}v_y$, $\mathrm{d}v_z$ are arbitrary, the only way for equation (12) to hold good is if all three terms vanish identically.

Therefore, for the v_x term, we would have

$$\begin{aligned} \frac{1}{f(v_x)}\frac{\partial f(v_x)}{\partial v_x} + 2Bv_x &= 0 \\ \implies \frac{\partial f(v_x)}{\partial v_x} &= -2Bv_x f(v_x) \\ \implies \frac{\partial f(v_x)}{f(v_x)} &= -2Bv_x \partial v_x \\ \implies \int \frac{\mathrm{d}f}{f} &= -2B \int v_x\,\mathrm{d}v_x \\ \implies \ln f(v_x) &= -2B\frac{v_x^2}{2} + \ln A \end{aligned}$$

where $\ln A$ is the constant of integration. From the above we obtain

$$\begin{aligned}
\ln f(v_x) - \ln A &= -Bv_x^2 \\
\implies \ln \frac{f(v_x)}{A} &= -Bv_x^2 \\
\implies \frac{f(v_x)}{A} &= e^{-Bv_x^2} \\
\implies f(v_x) &= Ae^{-Bv_x^2}
\end{aligned} \tag{13}$$

Likewise, from the v_y and v_z terms, we obtain

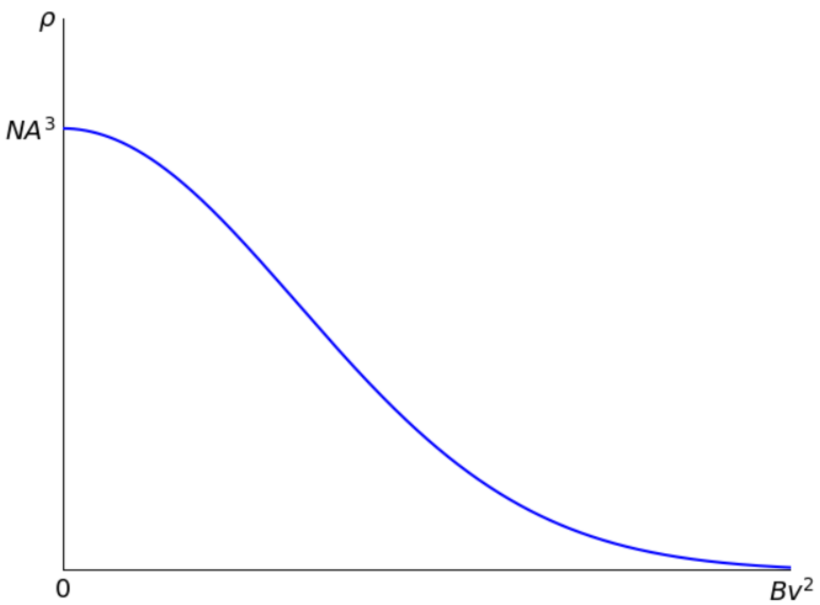
$$f(v_y) = Ae^{-Bv_y^2} \tag{14}$$

$$f(v_z) = Ae^{-Bv_z^2} \tag{15}$$

Using equations (13), (14) and (15) in equation (7) gives us

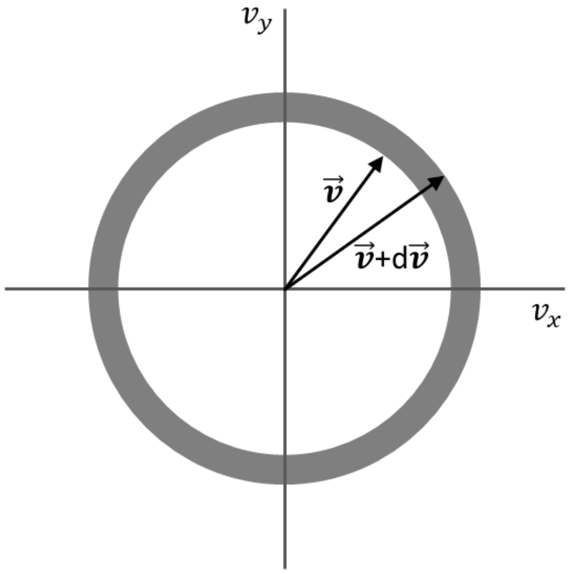
$$\begin{aligned}
\rho &= N\left(Ae^{-Bv_x^2}\right)\left(Ae^{-Bv_y^2}\right)\left(Ae^{-Bv_z^2}\right) \\
&= NA^3 e^{-B(v_x^2+v_y^2+v_z^2)} \\
\implies & \boxed{\rho(v) = NA^3 e^{-Bv^2}}
\end{aligned} \tag{16}$$

Equation (16) is known as *Maxwell's velocity density distribution function*, where ρ varies with molecular speeds v as shown in the plot below:



1.4.1 Molecular distribution of speeds

Consider a spherical shell of radius v and thickness $\mathrm{d}v$ in the velocity space. Then all points within this shell correspond to molecules with speeds in the range v and $v+\mathrm{d}v$. In the spherical



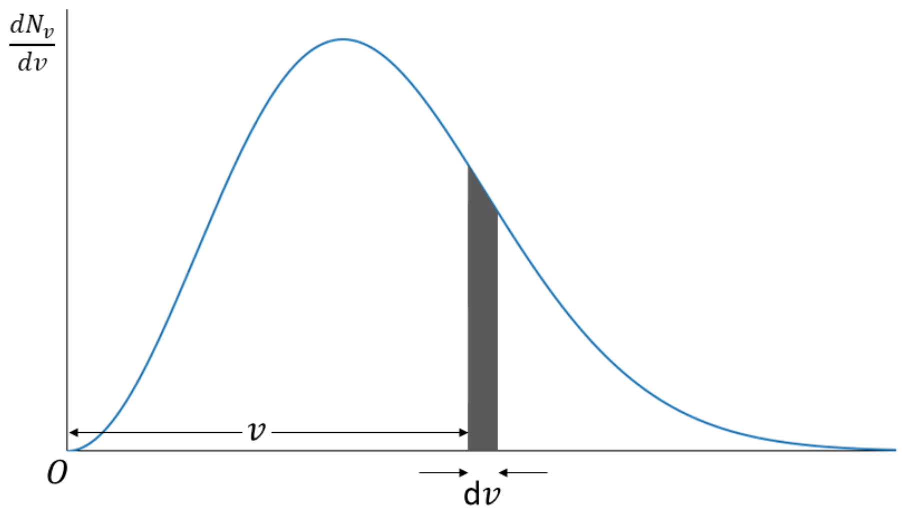
polar coordinates, the volume of an arbitrary volume element is

$$\mathrm{d}v_x\,\mathrm{d}v_y\,\mathrm{d}v_z = v^2 \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi\,\mathrm{d}v \tag{1}$$

The number of molecules with speeds in the range v and $v+\mathrm{d}v$ is obtained by integrating ρ over this shell as

$$\begin{aligned}
\mathrm{d}N_v &= \mathrm{d}^3 N_{v_x v_y v_z}\\
&= \int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi} N A^3 e^{-Bv^2} v^2 \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi\,\mathrm{d}v\\
&= N A^3 e^{-Bv^2} v^2\,\mathrm{d}v \int_0^{\pi} \sin\theta\,\mathrm{d}\theta \int_0^{2\pi} \mathrm{d}\phi\\
&= N A^3 e^{-Bv^2} v^2\,\mathrm{d}v\,[\cos 0 - \cos\pi] 2\pi\\
&= N A^3 e^{-Bv^2} v^2\,\mathrm{d}v \times 2 \times 2\pi\\
\Longrightarrow \mathrm{d}N_v &= 4\pi N A^3 v^2 e^{-Bv^2}\,\mathrm{d}v\\
\Longrightarrow \boxed{\frac{\mathrm{d}N_v}{\mathrm{d}v} = 4\pi N A^3 v^2 e^{-Bv^2}}
\end{aligned} \tag{2}$$

Equation (2) gives us the *Maxwellian distribution function of molecular speeds.*



In the plot of $\frac{\mathrm{d}N_v}{\mathrm{d}v}$ with respect to v, the area of a vertical shaded strip under the graph gives the number of molecules which have speeds between v and $v+\mathrm{d}v$. Hence, the total area under the entire curve gives us the total number of molecules N.

1.4.2 Determination of the constants

The total number of molecules N can be obtained by integrating $\mathrm{d}N_v$ over all speeds, i.e.

$$N = \int \mathrm{d}N_v = 4\pi N A^3 \int_0^\infty v^2 e^{-Bv^2}\,\mathrm{d}v \tag{1}$$

We now use the following standard integral:

$$\int_0^\infty e^{-\alpha x^2} x^n\,\mathrm{d}x = \frac{1}{2\alpha^{(n+1)/2}}\Gamma\left(\frac{n+1}{2}\right) \tag{2}$$

where the gamma function is defined as

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t}\,\mathrm{d}t$$

Some important values of $\Gamma(x)$ are given below:

x	$\Gamma(x)$
$1/2$	$\sqrt{\pi}$
1	1
$3/2$	$\sqrt{\pi}/2$
2	1

The gamma function also satisfies the recurrence relation: $\Gamma(n+1) = n\Gamma(n)$

In equation (2), setting $x \to v$, $\alpha \to B$ and $n = 2$ gives us

$$\int_0^\infty e^{-Bv^2} v^2 \,\mathrm{d}v = \frac{1}{2B^{(2+1)/2}} \Gamma\left(\frac{2+1}{2}\right)$$
$$= \frac{1}{2B^{3/2}} \Gamma\left(\frac{3}{2}\right)$$
$$\implies \int_0^\infty v^2 e^{-Bv^2} \,\mathrm{d}v = \frac{\sqrt{\pi}}{4B^{3/2}} \tag{3}$$

Inserting equation (3) in equation (1), we get

$$\cancel{N} = \cancel{4}\pi \cancel{N} A^3 \times \frac{\sqrt{\pi}}{\cancel{4}B^{3/2}}$$
$$\implies B^{3/2} = \sqrt{\pi^3} A^3$$
$$\implies A^3 = \frac{B^{3/2}}{\sqrt{\pi^3}} = \frac{B^{3/2}}{\pi^{3/2}} = \left(\sqrt{\frac{B}{\pi}}\right)^3$$
$$\implies A = \sqrt{\frac{B}{\pi}} \tag{4}$$

However, equation (4) only gives us the constant A in terms of another constant B. We can find B using the law of equipartition of energy of a gas.

The average value of v^2 is obtained as

$$\left\langle v^2 \right\rangle = \frac{\int_0^\infty v^2 \,\mathrm{d}N_v}{\int_0^\infty \mathrm{d}N_v} \tag{5}$$

Now, we have

$$\int_0^\infty v^2 \,\mathrm{d}N_v = 4\pi N A^3 \int_0^\infty v^4 e^{-Bv^2} \,\mathrm{d}v$$
$$= 4\pi N A^3 \frac{\Gamma(5/2)}{2B^{5/2}} \quad \text{[using the standard integral]}$$
$$\implies \int_0^\infty v^2 \,\mathrm{d}N_v = \frac{2\pi N A^3}{B^{5/2}} \Gamma\left(\frac{5}{2}\right)$$

Also, we have

$$\int_0^\infty \mathrm{d}N_v = 4\pi N A^3 \int_0^\infty v^2 e^{-Bv^2} \,\mathrm{d}v = \frac{2\pi N A^3}{B^{3/2}} \Gamma\left(\frac{3}{2}\right)$$

Using the above two integrals in equation (5), we obtain

$$
\begin{aligned}
\left\langle v^2\right\rangle &= \frac{\cancel{2\pi N A^3}\,\Gamma(5/2)}{B^{5/2}}\cdot\frac{B^{3/2}}{\cancel{2\pi N A^3}\,\Gamma(3/2)}\\
&= \frac{B^{3/2}}{B^{5/2}}\cdot\frac{\Gamma(5/2)}{\Gamma(3/2)}\\
&= \frac{B^{3/2}}{B^{5/2}}\cdot\frac{{}^3\!/_2\,\cancel{\Gamma(3/2)}}{\cancel{\Gamma(3/2)}}\\
&= \frac{3}{2}B^{3/2-5/2} = \frac{3}{2}B^{-1}\\
\Longrightarrow \left\langle v^2\right\rangle &= \frac{3}{2B}
\end{aligned}
\tag{6}
$$

Hence, the average kinetic energy of the gas molecules becomes

$$
\begin{aligned}
\langle K\rangle &= \frac{1}{2}m\left\langle v^2\right\rangle = \frac{1}{2}m\times\frac{3}{2B}\\
\Longrightarrow \langle K\rangle &= \frac{3m}{4B}
\end{aligned}
\tag{7}
$$

However, from the law of equipartition of energy of a gas, the average kinetic energy of the gas molecules at temperature T is

$$
\langle K\rangle = \frac{3}{2}k_B T \tag{8}
$$

where $k_B = 1.381\times10^{-23}\,\mathrm{J\,K^{-1}}$ is the Boltzmann constant.

Comparing equations (7) and (8) gives us

$$
\begin{aligned}
\frac{3m}{4B} &= \frac{3}{2}k_B T\\
\Longrightarrow \frac{\cancel{3}m}{\cancel{4}^{\,2}\,B} &= \frac{\cancel{3}}{\cancel{2}}k_B T\\
\Longrightarrow B &= \frac{m}{2k_B T}
\end{aligned}
\tag{9}
$$

Inserting equation (9) in equation (4) gives us the value of the constant A:

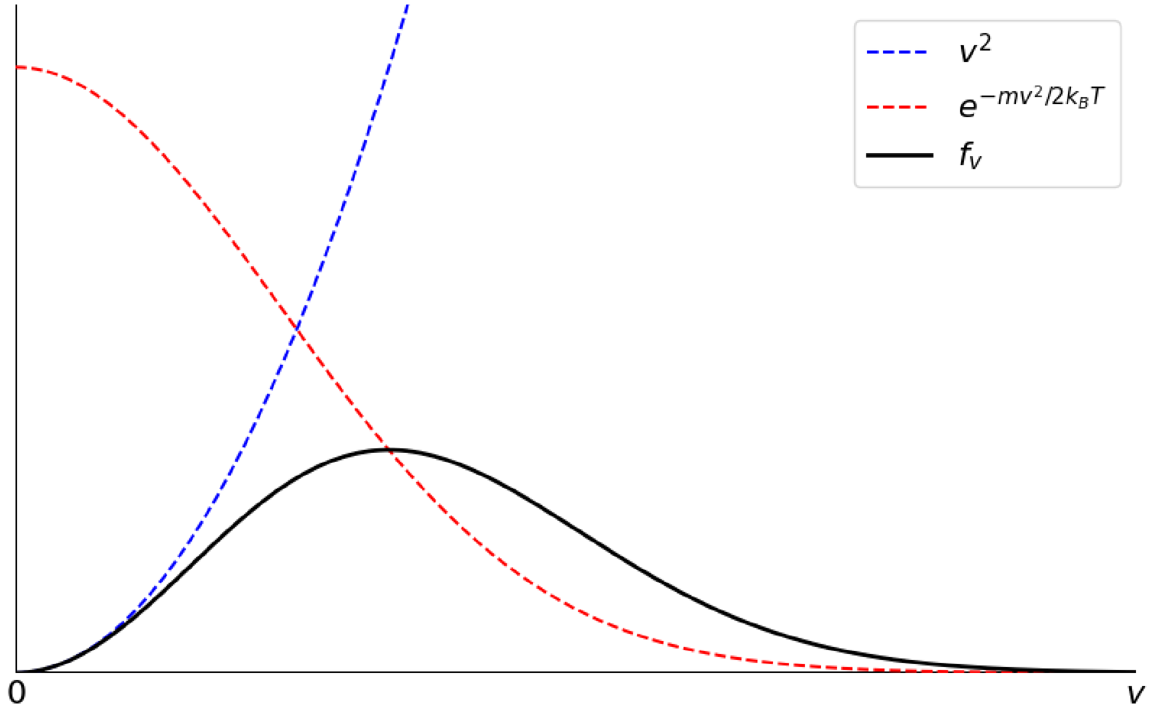
$$
\begin{aligned}
A &= \sqrt{\frac{B}{\pi}}\\
\Longrightarrow A &= \sqrt{\frac{m}{2\pi k_B T}}
\end{aligned}
\tag{10}
$$

Finally, substituting for the constants A and B using equations (10) and (9) respectively in the Maxwellian distribution function for molecular speeds, we get

$$
\begin{aligned}
\mathrm{d}N_v &= 4\pi N\left(\frac{m}{2\pi k_B T}\right)^{3/2} v^2 e^{-mv^2/2k_B T}\,\mathrm{d}v\\
\Longrightarrow &\boxed{f_v = \frac{\mathrm{d}N_v}{N\,\mathrm{d}v} = 4\pi\left(\frac{m}{2\pi k_B T}\right)^{3/2} v^2 e^{-mv^2/2k_B T}}
\end{aligned}
\tag{11}
$$

The *Maxwellian density function* f_v, from equation (11), gives us the probability per unit molecular speed of finding a molecule with speed between v and $v+\mathrm{d}v$. This function has the following features:

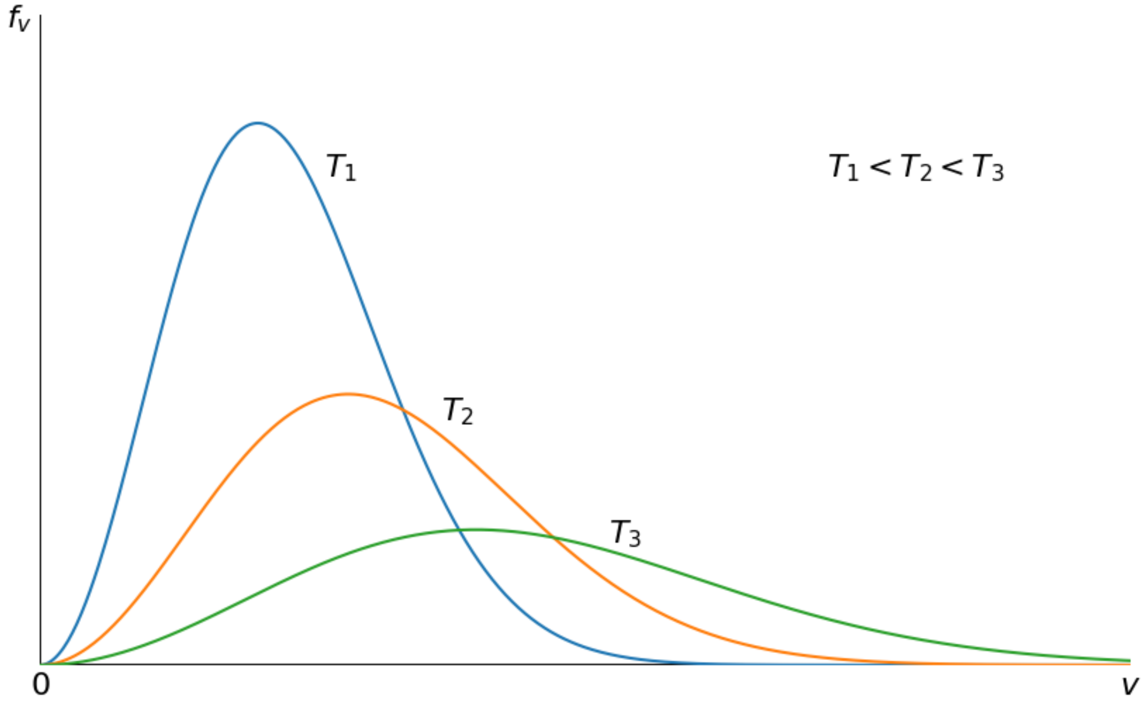
1. $f_v = 0$ for $v = 0$ (due to v^2) and $f_v \to 0$ for $v \to \infty$ (due to $e^{-mv^2/2k_BT}$).
2. For small molecular speeds, the v^2 factor dominates and f_v increases quadratically.
3. As the speed increases, the $e^{-mv^2/2k_BT}$ factor begins to dominate and the distribution starts decaying exponentially after attaining a maximum value.



4. Because f_v is a probability density function, at all T the following condition must hold good:

$$\int_0^\infty f_v \,\mathrm{d}v = 1$$

5. As the temperature T increases, more and more molecules gain kinetic energy and so the peak shifts towards higher molecular speeds. But the total number of molecules remains constant, and so to keep the corresponding area under the curve constant, curve becomes progressively flatter with increasing T.



Exercises

1. Calculate the probability that the speed of an O_2 molecule will lie between 100 m/s and 101 m/s at 200 K. The mass of an O_2 molecule is 32 u. Take $k_B = 1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}}$ and $N_A = 6.023 \times 10^{26}\,\mathrm{kmol^{-1}}$.

2. A gas composed of 10^6 carbon atoms has a Maxwellian velocity distribution at $T = 300\,\mathrm{K}$. Determine the number of atoms having velocities between
   (a) $100\,\mathrm{m\,s^{-1}}$ and $101\,\mathrm{m\,s^{-1}}$
   (b) $300\,\mathrm{m\,s^{-1}}$ and $301\,\mathrm{m\,s^{-1}}$
   (c) $1500\,\mathrm{m\,s^{-1}}$ and $1501\,\mathrm{m\,s^{-1}}$

2 Molecular speeds

2.1 Average or mean speed

The average speed of the gas is defined as

$$v_{\text{avg}} = \langle v \rangle = \frac{1}{N} \int_0^\infty v \, \mathrm{d}N_v$$

Substituting for $\mathrm{d}N_v$ from the Maxwellian density function, we have

$$v_{\text{avg}} = \frac{1}{N} \times 4\pi N \left(\frac{m}{2\pi k_B T} \right)^{3/2} \underbrace{\int_0^\infty v^3 e^{-mv^2/2k_B T} \, \mathrm{d}v}_{= \frac{1}{2(m/2k_B T)^2}}$$

$$\begin{aligned}
&= \cancelto{2}{4}\, \pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \frac{1}{\cancel{2}(m/2k_B T)^2} \\
&= 2\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \left(\frac{2k_B T}{m} \right)^2 \\
&= \frac{2\pi^{1-3/2} m^{3/2-2}}{2^{3/2-2} k_B^{3/2-2} T^{3/2-2}} = \frac{2^{3/2} k_B^{1/2} T^{1/2}}{\pi^{1/2} m^{1/2}} \\
&= \left(\frac{2^3 k_B T}{\pi m} \right)^{1/2}
\end{aligned}$$

$$\implies \boxed{v_{\text{avg}} = \sqrt{\frac{8k_B T}{\pi m}}}$$

2.2 Root mean squared (RMS) speed

The root mean squared (RMS) speed is defined as

$$v_{\text{rms}} = \sqrt{\langle v^2 \rangle}$$

Now, we already have that

$$\left\langle v^2 \right\rangle = \frac{3}{2B} \quad \text{and} \quad B = \frac{m}{2k_B T}$$

The above give us

$$\left\langle v^2 \right\rangle = \frac{3}{\cancel{2}} \times \frac{\cancel{2}k_B T}{m} = \frac{3k_B T}{m}$$

Therefore the RMS speed is

$$\boxed{v_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}}$$

2.3 Most probable speed

The most probable speed v_p is defined as the speed at which the probability of finding a molecule is maximum.

- This is obtained by finding the root of the equation $\frac{\mathrm{d}f_v}{\mathrm{d}v} = 0$.

We have

$$f_v = 4\pi\left(\frac{m}{2\pi k_B T}\right)^{3/2} v^2 e^{-mv^2/2k_B T} \tag{1}$$

Differentiating equation (1) with respect to v and equating to 0 gives $v = v_p$, we have

$$\begin{aligned}
\frac{\mathrm{d}f_v}{\mathrm{d}v} &= 0 \\
\implies 4\pi\left(\frac{m}{2\pi k_B T}\right)^{3/2} \frac{\mathrm{d}}{\mathrm{d}v}\left[v^2 e^{-mv^2/2k_B T}\right] &= 0 \\
\implies \frac{\mathrm{d}}{\mathrm{d}v}\left[v^2 e^{-mv^2/2k_B T}\right] &= 0 \\
\implies 2v_p e^{-mv^2/2k_B T} + v_p^2\left(-\frac{m}{2k_B T}\right)2v_p e^{-mv^2/2k_B T} &= 0 \\
\implies 2v_p e^{-mv^2/2k_B T}\left[1 - \frac{mv_p^2}{2k_B T}\right] &= 0
\end{aligned}$$

On the LHS of the above, the factor $e^{-mv^2/2k_B T}$ cannot be zero for any finite value of v. Hence we must have

$$\begin{aligned}
1 - \frac{mv_p^2}{2k_B T} &= 0 \\
\implies \frac{mv_p^2}{2k_B T} &= 1 \\
\implies v_p^2 &= \frac{2k_B T}{m} \\
\implies & \boxed{v_p = \sqrt{\frac{2k_B T}{m}}}
\end{aligned}$$

Summary

The three molecular speeds are as follows:

$$v_{\text{avg}} = \sqrt{\frac{8k_B T}{\pi m}}, \qquad v_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}, \qquad v_p = \sqrt{\frac{2k_B T}{m}}$$

This gives their ratios to be

$$v_{\text{avg}} : v_{\text{rms}} : v_p = \sqrt{\frac{8}{\pi}} : \sqrt{3} : \sqrt{2} \approx 1.6 : 1.7 : 1.4$$

The above shows that

$$v_p < v_{\text{avg}} < v_{\text{rms}}$$

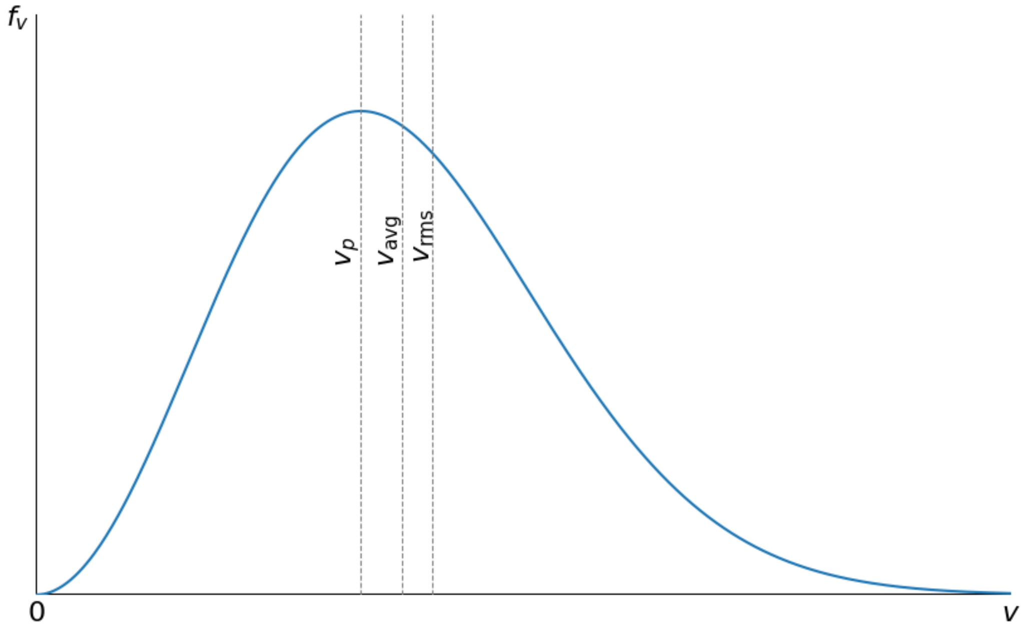


Table 1: Table showing values of v_{avg}, v_{rms} and v_p for different gases at STP

Gas	$v_{\text{avg}}\,(\mathrm{m\,s^{-1}})$	$v_{\text{rms}}\,(\mathrm{m\,s^{-1}})$	$v_p\,(\mathrm{m\,s^{-1}})$
H_2	1695	1838	1501
H_2O	567	615	502
N_2	455	493	403
Air	447	485	396
O_2	425	461	376
CO_2	362	393	321

Exercises

1. For a Maxwellian gas, show that $\langle v\rangle \left\langle \frac{1}{v} \right\rangle = \frac{4}{\pi}$.

2. Calculate the most probable speed, average speed and root mean square speed for O_2 molecules at 300 K using the following data:
$$m(\mathrm{O_2}) = 5.31\times10^{-26}\,\mathrm{kg} \quad \text{and} \quad k_B = 1.38\times10^{-23}\,\mathrm{J\,K^{-1}}$$

3. Calculate the fraction of molecules of a gas within 1% of the most probable speed at STP. Will this fraction remain the same at all temperatures?

3 Energy distribution of a Maxwellian gas

In order the derive the distribution of energies of a Maxwellian gas, the following assumptions are made:

1. The energy of a Maxwellian gas is almost entirely due to the kinetic energy, i.e. $E = K$.
2. The number of molecules with kinetic energy in the range E and $E + \mathrm{d}E$ is the same as the number of molecules with speeds in the range v and $v + \mathrm{d}v$.

We know that for a molecule with mass m and moving with a speed v, its kinetic energy is

$$
\begin{aligned}
E &= \frac{1}{2} m v^2 \\
\implies v^2 &= \frac{2E}{m} && (1) \\
\implies v &= \sqrt{\frac{2E}{m}} && (2)
\end{aligned}
$$

Taking the differential of equation (1), we have

$$
\begin{aligned}
2v\,\mathrm{d}v &= \frac{2\,\mathrm{d}E}{m} \\
\implies \mathrm{d}E &= m v\,\mathrm{d}v \\
&= m\sqrt{\frac{2E}{m}}\,\mathrm{d}v \quad \text{[using eqn. (2)]} \\
\implies \mathrm{d}E &= \sqrt{2mE}\,\mathrm{d}v && (3)
\end{aligned}
$$

Now, from assumption #2, we have

$$
\begin{aligned}
\mathrm{d}N_E &= \mathrm{d}N_v \\
\implies \mathrm{d}N_E &= 4\pi N \left(\frac{m}{2\pi k_B T}\right)^{3/2} v^2 \exp\left[-\frac{m v^2}{2k_B T}\right] \mathrm{d}v \\
&= 4\pi N \left(\frac{m}{2\pi k_B T}\right)^{3/2} \underbrace{\left(\frac{2E}{m}\right)}_{\text{from eqn. (1)}} \underbrace{\exp\left[-\frac{E}{k_B T}\right]}_{\because E = \frac{m v^2}{2}} \underbrace{\left(\frac{\mathrm{d}E}{\sqrt{2mE}}\right)}_{\text{from eqn. (3)}} \\
&= 4\pi N \frac{m^{3/2} 2E m^{-1}}{2\sqrt{2}\pi^{3/2}(k_B T)^{3/2}} \cdot \frac{1}{\sqrt{2} m^{1/2} E^{1/2}} \exp\left[-\frac{E}{k_B T}\right] \mathrm{d}E \\
&= \frac{2\pi^{1-3/2} N m^{3/2-1-1/2} E^{1-1/2}}{(k_B T)^{3/2}} \exp\left[-\frac{E}{k_B T}\right] \mathrm{d}E \\
&= \frac{2N\pi^{-1/2} E^{1/2}}{(k_B T)^{3/2}} \exp\left[-\frac{E}{k_B T}\right] \mathrm{d}E \\
\implies & \boxed{\mathrm{d}N_E = 2N\left(\frac{E}{\pi}\right)^{1/2} \left(\frac{1}{k_B T}\right)^{3/2} \exp\left[-\frac{E}{k_B T}\right] \mathrm{d}E} && (4)
\end{aligned}
$$

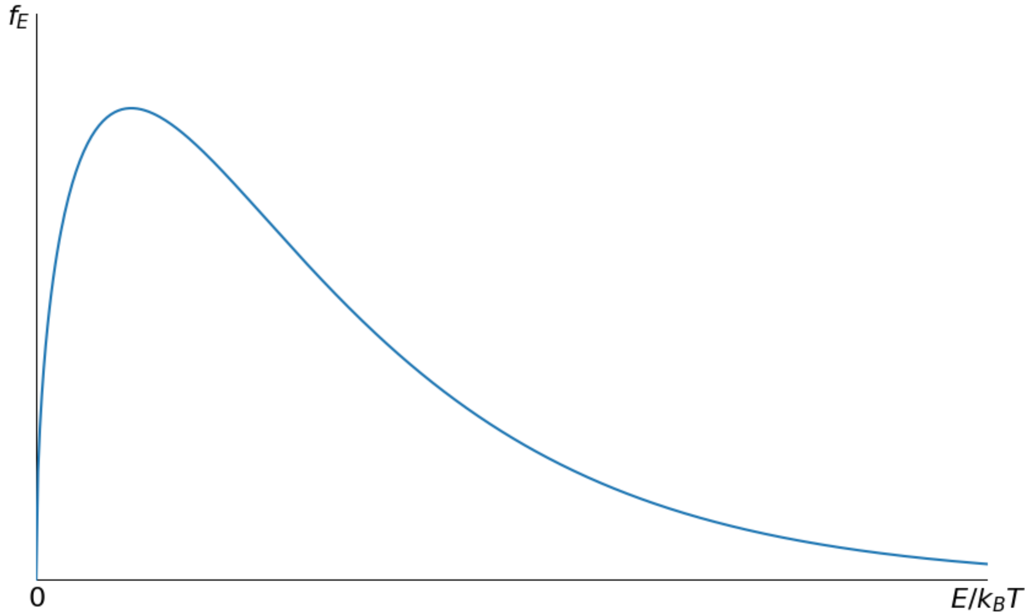
Equation (4) gives us the number of molecules with energies between E and $E + \mathrm{d}E$.

Further, from equation (4), we find the probability of finding a molecule with energies between E and $E + \mathrm{d}E$ to be

$$\boxed{\mathrm{d}P_E = \frac{\mathrm{d}N_E}{N} = 2\left(\frac{E}{\pi}\right)^{1/2}\left(\frac{1}{k_B T}\right)^{3/2} \exp\left[-\frac{E}{k_B T}\right]\mathrm{d}E} \tag{5}$$

Finally, from equation (5), we obtain the probability density function for molecular energies to be

$$\boxed{f_E = \frac{\mathrm{d}P_E}{\mathrm{d}E} = \frac{\mathrm{d}N_E}{N\,\mathrm{d}E} = 2\left(\frac{E}{\pi}\right)^{1/2}\left(\frac{1}{k_B T}\right)^{3/2} \exp\left[-\frac{E}{k_B T}\right]} \tag{6}$$



4 Molecular energies

4.1 Average energy

The average kinetic energy of a molecule is given as

$$
\begin{aligned}
\varepsilon_{\text{avg}} = \langle E \rangle &= \frac{1}{N}\int_0^\infty E\,\mathrm{d}N_E \\
&= \frac{1}{\cancel{N}} \times 2\cancel{N}\frac{1}{\sqrt{\pi}}\left(\frac{1}{k_B T}\right)^{3/2}\int_0^\infty E^{3/2}\exp\left[-\frac{E}{k_B T}\right]\mathrm{d}E \\
&= \frac{2}{\sqrt{\pi}(k_B T)^{3/2}}\int_0^\infty E^{3/2}\exp\left[-\frac{E}{k_B T}\right]\mathrm{d}E
\end{aligned}
$$

In order to evaluate the integral on the RHS above, we make the substitution:

$$
\begin{aligned}
x &= \frac{E}{k_B T} \\
\implies E &= (k_B T)x
\end{aligned}
$$

Also we have

$$
\begin{aligned}
\mathrm{d}x &= \frac{\mathrm{d}E}{k_B T} \\
\implies \mathrm{d}E &= k_B T\,\mathrm{d}x
\end{aligned}
$$

Then we can re-write the integral as

$$
\begin{aligned}
\int_0^\infty E^{3/2}\exp\left[-\frac{E}{k_B T}\right]\mathrm{d}E &= \int_0^\infty (k_B T)^{3/2}x^{3/2}e^{-x}k_B T\,\mathrm{d}x \\
&= (k_B T)^{3/2}(k_B T)\underbrace{\int_0^\infty x^{3/2}e^{-x}\,\mathrm{d}x}_{=\Gamma\left(\frac{5}{2}\right)} \\
&= (k_B T)^{3/2}(k_B T)\Gamma\left(\frac{5}{2}\right) \\
\implies \int_0^\infty E^{3/2}\exp\left[-\frac{E}{k_B T}\right]\mathrm{d}E &= (k_B T)^{3/2}(k_B T)\frac{3\sqrt{\pi}}{4}
\end{aligned}
$$

This gives us

$$
\begin{aligned}
\varepsilon_{\text{avg}} &= \frac{\cancel{2}}{\cancel{\sqrt{\pi}}\cancel{(k_B T)^{3/2}}}\cancel{(k_B T)^{3/2}}(k_B T)\frac{3\cancel{\sqrt{\pi}}}{\cancel{4}_{2}} \\
\implies &\boxed{\varepsilon_{\text{avg}} = \frac{3}{2}k_B T}
\end{aligned}
$$

4.2 Most probable energy

To obtain the most probable energy ε_p, we need to find the energy such that $\dfrac{\mathrm{d}f_E}{\mathrm{d}E} = 0$.

Therefore,

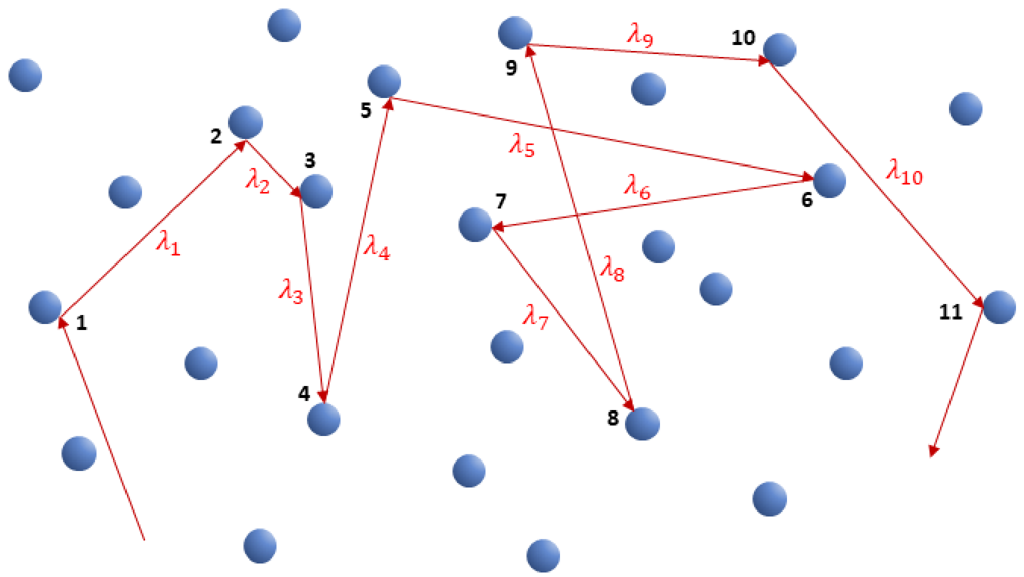
$$
\begin{aligned}
\frac{\mathrm{d}f_E}{\mathrm{d}E} &= 0 \\
\implies \frac{\mathrm{d}}{\mathrm{d}E}\left[2\left(\frac{E}{\pi}\right)^{1/2}\left(\frac{1}{k_B T}\right)^{3/2}\exp\left(-\frac{E}{k_B T}\right)\right] &= 0 \\
\implies \frac{2}{\sqrt{\pi}(k_B T)^{3/2}}\frac{\mathrm{d}}{\mathrm{d}E}\left[E^{1/2}\exp\left(-\frac{E}{k_B T}\right)\right] &= 0 \\
\implies \frac{\mathrm{d}}{\mathrm{d}E}\left[E^{1/2}\exp\left(-\frac{E}{k_B T}\right)\right] &= 0 \\
\implies \frac{1}{2\varepsilon_p^{1/2}}\cancel{\exp\left(-\frac{E}{k_B T}\right)} + \varepsilon_p^{1/2}\left(-\frac{1}{k_B T}\right)\cancel{\exp\left(-\frac{E}{k_B T}\right)} &= 0 \\
\implies \frac{1}{2\varepsilon_p^{1/2}} - \frac{\varepsilon_p^{1/2}}{k_B T} &= 0 \\
\implies \frac{\varepsilon_p^{1/2}}{k_B T} &= \frac{1}{2\varepsilon_p^{1/2}} \\
\implies \boxed{\varepsilon_p^{1/2} = \frac{1}{2}k_B T}
\end{aligned}
$$

- We find that $\varepsilon_{\text{avg}} = 3\varepsilon_p$.
- At any given temperature, ε_{avg} and ε_p are the same for all gases.

5 Molecular collisions

5.1 Mean free path

A gas attains thermal equilibrium after successive molecular collisions. We define *free path* as the distance covered by a particle between successive collisions. In the figure below, the paths taken by a gas molecule between collisions is shown. For instance, λ_1 is the free path between collisions 1 and 2. Similarly λ_2 is the free path between collisions 2 and 3, and so on.



The *mean free path* λ of a gas is the average distance travelled by a molecule between two successive collisions. Suppose a molecule undergoes N collisions, then we have

$$\begin{aligned} \lambda &= \frac{\text{total distance travelled}}{\text{total number of collisions}} \\ &= \frac{\lambda_1 + \lambda_2 + \cdots + \lambda_N}{N} \\ \implies \lambda &= \frac{1}{N}\sum_{i=1}^{N}\lambda_i \end{aligned} \tag{1}$$

If T is the total time taken by a molecule to undergo N collisions and $\langle v \rangle$ is the average molecular speed, then we have

$$\sum_{i=1}^{N}\lambda_i = \langle v \rangle\, T$$

Then equation (1) becomes

$$\lambda = \frac{\langle v \rangle\, T}{N} \tag{2}$$

Now, the mean time between two successive collisions can be obtained as

$$\tau = \frac{T}{N}$$

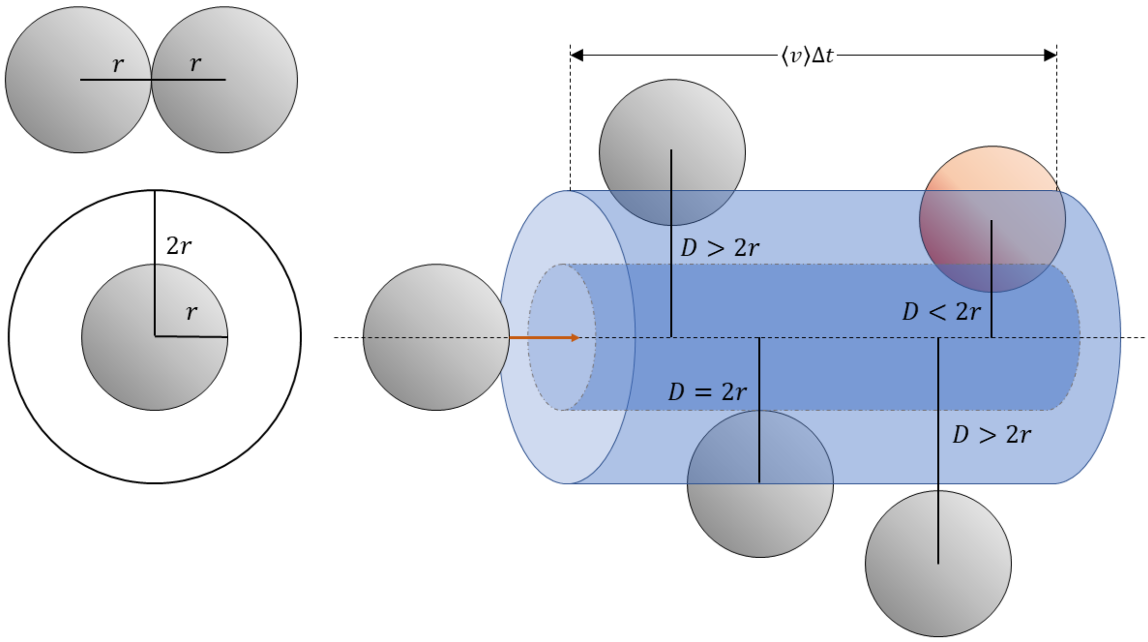
Then, equation (2) becomes

$$\lambda = \langle v \rangle \tau = \frac{\langle v \rangle}{P_c} \tag{3}$$

where the collision frequency $P_c = \dfrac{1}{\tau}$ measures the average number of collisions per unit time.

5.1.1 Estimation of λ

I. *Zeroth order approximation*: In this approximation, it is assumed that all molecules of the gas are immobile, except for one.



If r is the radius of a molecule, then for a collusion to occur, the centre-to-centre distance between the moving and stationary molecules would be equal to its diameter $d = 2r$.

We may visualize this as if the moving molecule carries with it a circular disc of radius d. Therefore, in time Δt, it can be thought of as sweeping out a cylinder of cross sectional area πd^2 and length $\langle v \rangle \Delta t$.

Collisions would occur only with molecules whose centres lie within this cylinder.

We define the microscopic collision cross-section σ as

$$\sigma = \pi d^2$$

During the time interval Δt, the molecule would collide with all other molecules whose centres lie within the cylinder of volume

$$\Delta V = \pi d^2 \langle v \rangle \Delta t$$

If n is the number density of the gas, then the total number of molecules contained within the volume ΔV is

$$\Delta N = n\Delta V$$
$$\implies \Delta N = n\pi d^2 \langle v \rangle \Delta t$$

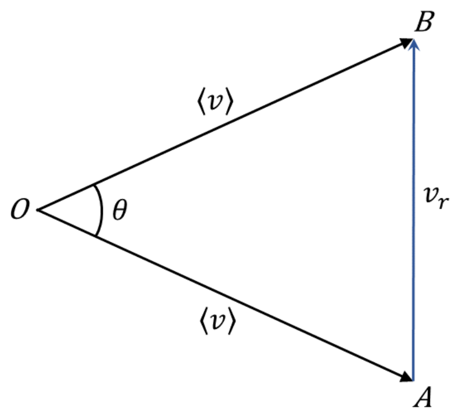
But ΔN above is also the number of collisions experienced by the moving molecule in time Δt. So we can obtain the collision frequency as

$$P_c = \frac{\Delta N}{\Delta t} = n\pi d^2 \langle v \rangle$$
$$\implies P_c = n\sigma \langle v \rangle$$

This gives us the mean free path to be

$$\lambda = \frac{\langle v \rangle}{P_c} = \frac{\cancel{\langle v \rangle}}{n\sigma \cancel{\langle v \rangle}}$$
$$\implies \boxed{\lambda = \frac{1}{n\sigma}}$$

II. *First order approximation*: In this approximation, the motion of all the molecules are taken into account.



Consider two molecules moving with average speed $\langle v \rangle$ along OA and OB with angle θ between them. Let v_r be the relative speed between the molecules. Then we have

$$v_r = \left[\langle v \rangle^2 + \langle v \rangle^2 - 2\langle v \rangle^2 \cos\theta\right]^{1/2}$$
$$\implies v_r = \left[2\langle v \rangle^2 (1-\cos\theta)\right]^{1/2} \tag{1}$$

We know that

$$\cos 2\theta = 1 - 2\sin^2\theta$$
$$\implies 1 - \cos 2\theta = 2\sin^2\theta$$
$$\implies 1 - \cos\theta = 2\sin^2\left(\frac{\theta}{2}\right) \quad [\text{Setting } \theta \to \theta/2]$$

Therefore, equation (1) becomes

$$v_r = \left[2\langle v \rangle^2 2\sin^2(\theta/2)\right]^{1/2}$$
$$\implies v_r = 2\langle v \rangle \sin\left(\frac{\theta}{2}\right) \tag{2}$$

To find the average value of v_r, we integrate equation (2) over all solid angles, i.e.

$$
\begin{aligned}
\langle v_r \rangle &= \frac{1}{4\pi} \int_0^{4\pi} v_r \, \mathrm{d}\Omega \\
&= \frac{1}{4\pi} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} v_r \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi \quad [\because \ \mathrm{d}\Omega = \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi] \\
&= \frac{1}{4\pi} \left[\int_0^{2\pi} \mathrm{d}\phi \right] \int_0^{\pi} v_r \sin\theta \, \mathrm{d}\theta \\
&= \frac{1}{\cancel{4\pi}_{2}} \times \cancel{2\pi} \int_0^{\pi} v_r \sin\theta \, \mathrm{d}\theta \\
&= \frac{1}{\cancel{2}} \int_0^{\pi} \left[\cancel{2} \langle v \rangle \sin\left(\frac{\theta}{2}\right) \right] \sin\theta \, \mathrm{d}\theta \quad [\text{Using eqn. (2)}] \\
&= \langle v \rangle \int_0^{\pi} \sin\theta \sin\frac{\theta}{2} \, \mathrm{d}\theta \\
&= \langle v \rangle \int_0^{\pi} \left(2\sin\frac{\theta}{2} \cos\frac{\theta}{2} \right) \sin\frac{\theta}{2} \, \mathrm{d}\theta \\
\implies \langle v_r \rangle &= 2 \langle v \rangle \int_0^{\pi} \sin^2\frac{\theta}{2} \cos\frac{\theta}{2} \, \mathrm{d}\theta
\end{aligned}
\tag{3}
$$

To evaluate the integral on the RHS of equation (3), we make the substitution:

$$
\begin{aligned}
x &= \sin\frac{\theta}{2} \\
\implies \mathrm{d}x &= \frac{1}{2} \cos\frac{\theta}{2} \, \mathrm{d}\theta \\
\implies \cos\frac{\theta}{2} \, \mathrm{d}\theta &= 2 \, \mathrm{d}x
\end{aligned}
$$

Then for the limits, we have

At $\theta = 0$, we have

$$x = \sin 0 = 0$$

At $\theta = \pi$, we have

$$x = \sin\frac{\pi}{2} = 1$$

Then we have

$$
\begin{aligned}
\int_0^{\pi} \sin^2\frac{\theta}{2} \cos\frac{\theta}{2} \, \mathrm{d}\theta &= \int_0^{1} \left(x^2 \right) (2 \, \mathrm{d}x) \\
&= 2 \int_0^{1} x^2 \, \mathrm{d}x = 2 \left. \frac{x^3}{3} \right|_0^1 = \frac{2}{3}
\end{aligned}
$$

Hence, from equation (3), we get

$$\begin{aligned} \langle v_r \rangle &= 2 \langle v \rangle \times \frac{2}{3} \\ \implies \langle v_r \rangle &= \frac{4}{3} \langle v \rangle \end{aligned}$$

Then the collision frequency gets modified from that in the zeroth order approximation as

$$\begin{aligned} P_c &= n\sigma \langle v_r \rangle \\ \implies P_c &= \frac{4}{3} N\sigma \langle v \rangle \end{aligned} \tag{4}$$

Equation (4) shows that when we take into account the motion of the other molecules, the collision frequency increases.

This also gives us the modified mean free path (or *Clausius mean free path*) to be

$$\begin{aligned} \lambda_{\mathrm{Cl}} &= \frac{\langle v \rangle}{P_c} \\ &= \frac{\cancel{\langle v \rangle}}{{}^4/_3 N\sigma \cancel{\langle v \rangle}} \\ &= \frac{3}{4N\sigma} \\ \implies \lambda_{\mathrm{Cl}} &= \frac{0.75}{n\sigma} \end{aligned} \tag{5}$$

As expected, equation (5) shows that when the motion of the other molecules are taken into account, the mean free path decreases.